

An Approach to Pell's Equation via Algebraic Number Theory

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0 Introduction

This paper is an exposition of an approach to Pell's Equation that illuminates the equation's deep connections with algebraic number theory. This paper draws material from a few sources. Most proofs presented are adapted from Pierre Samuel's 1970 text *Algebraic Theory of Numbers* [3], though I have taken significant effort to reword and expand these proofs in order to give clarity to the subject. Other references include David Cox's 2012 text *Galois Theory* [1] and Frazer Jarvis' 2014 text *Algebraic Number Theory* [2]. The main aim of this paper is to give clear, detailed proofs while demonstrating a link between apparently disparate subjects. Throughout, I assume that the reader is comfortable with the various algebraic structures discussed in a first course on algebra.

1 Pell's Equation

Pell's Equation is a particular Diophantine equation that has received much attention over the past centuries. If we fix some $d \in \mathbb{N}$, then a pair of integers (x, y) is said to satisfy Pell's Equation if

$$x^2 - dy^2 = 1.$$

The first thing to see is that a restriction can be placed on d without losing generality. Suppose there is some prime p such that $p^2 | d$. Then a pair of integers (x_0, y_0) is a solution to $x^2 - dy^2 = 1$ if and only if the pair (x_0, py_0) is a solution to $x^2 - \frac{d}{p^2}y^2 = 1$. We therefore only study Pell's Equation in the case where d is not divisible by any square, since solutions for arbitrary d can be determined by this special case. A number $d \in \mathbb{N}$ is said to be **squarefree** if there is no prime p such that $p^2 | d$.

An elementary application of Pell's Equation is the approximation of square roots. Suppose that (x, y) satisfies Pell's Equation. Then we may rearrange

$$\begin{aligned} x^2 - dy^2 &= 1, \\ x^2 &= 1 + dy^2, \\ \left(\frac{x}{y}\right)^2 &= \frac{1}{y^2} + d. \end{aligned}$$

Therefore the rational number $\frac{x}{y}$ is an approximation to $\sqrt{d}$. If we are able to find solutions $(x_0, y_0), (x_1, y_1), \dots$ to Pell's Equation such that the sequence of y_i 's is unbounded, then $\frac{1}{y_i^2} \rightarrow 0$ as $i \rightarrow \infty$, so we can approximate $\sqrt{d}$ to arbitrary precision. In Section 5 we will see an algorithm that allows us to generate such a sequence of solutions. For now, we turn to the theory of number fields, a special case of which is intimately related to Pell's Equation.

2 Number Fields

We first take a brief survey of some concepts basic to the algebraic study of number theory. The definitions and results given here can be found in more detail in Cox's *Galois Theory* [1]

2.1 Basic Concepts

If L is a field, and F is a subfield of L , then we say that L is a **field extension** of F . A field extension L of F can be viewed as vector space over F , and the dimension of this vector space is called the degree of the extension, which is written $[L : F]$. If L is a field extension of F and $\alpha \in L$ such that α satisfies a polynomial over F , we say that α is **algebraic** over F . The monic polynomial over F of smallest degree that α satisfies is called the **minimal polynomial** of α , and this polynomial is necessarily irreducible over $\mathbb{Q}$. If $\alpha_1, \dots, \alpha_n$ are algebraic over F , then we define the field

$$F(\alpha_1, \dots, \alpha_n) := \{f(\alpha_1, \dots, \alpha_n)/g(\alpha_1, \dots, \alpha_n) \mid f, g \in F[x_1, \dots, x_n]\},$$

where $F[x_1, \dots, x_n]$ denotes the ring of n -variable polynomials over F . A finite-degree extension K of $\mathbb{Q}$ is called a **number field**. It is an elementary result that if α is algebraic over a field F , then the degree of $F(\alpha)$ over F is precisely the degree of the minimal polynomial of α . Another fundamental result in the theory of field extensions is that, if F is of characteristic 0, then every field extension L of F is generated by a **primitive element**. That is, if L is an extension, then there is some $\alpha \in L$ such that $L = F(\alpha)$. Finally, if K and L are extensions of F such that L is an extension K , then the **Tower Theorem** states that

$$[L : F] = [L : K] \cdot [K : F].$$

2.2 Quadratic Fields

This subsection draws material from [3, §2.5].

A number field K of degree 2 over $\mathbb{Q}$ is called a **quadratic field**. By the preceding section, we see that there is some $\alpha \in K$ with a quadratic minimal polynomial such that $K = \mathbb{Q}(\alpha)$. However, even more can be said in the case of quadratic fields. If β is an element of K such that $\beta \notin \mathbb{Q}$, then β certainly satisfies no linear polynomial. Therefore the minimal polynomial of β over $\mathbb{Q}$ has degree at least 2. From the Tower Theorem it follows that

$$2 = [K : \mathbb{Q}] = [K : \mathbb{Q}(\beta)] \cdot [\mathbb{Q}(\beta) : \mathbb{Q}] \geq [K : \mathbb{Q}(\beta)] \cdot 2.$$

Therefore K must have dimension 1 as a $\mathbb{Q}(\beta)$ -vector space, and so $K = \mathbb{Q}(\beta)$. Therefore every $\beta \in K \setminus \mathbb{Q}$ generates K . It turns out that quadratic fields admit of a particularly nice representation.

Theorem 1. *If K is a quadratic field, then there is some squarefree integer $d \in \mathbb{Z}$ such that $K = \mathbb{Q}(\sqrt{d})$.*

Proof. Let $K = \mathbb{Q}(\alpha)$ be a quadratic field, and let $f(x) = x^2 + bx + c$ be the minimal polynomial of α over $\mathbb{Q}$. By the quadratic formula, we have that $2\alpha + b = \pm\sqrt{b^2 - 4c}$. From $\alpha \notin \mathbb{Q}$ it follows that $2\alpha + b \notin \mathbb{Q}$, and so $\sqrt{b^2 - 4c} \notin \mathbb{Q}$. It therefore follows that $K = \mathbb{Q}(\sqrt{b^2 - 4c})$. Since $b^2 - 4c$ is in $\mathbb{Q}$, there are integers u and v such that $b^2 - 4c = \frac{u}{v} = \frac{uv}{v^2}$. Therefore $\frac{\sqrt{uv}}{v} \notin \mathbb{Q}$, and so we must have $\sqrt{uv} \notin \mathbb{Q}$, so that $K = \mathbb{Q}(\sqrt{uv})$. Finally, if uv is not squarefree, then let d be the greatest squarefree integer dividing uv . It is easy to see that $\sqrt{uv} = k\sqrt{d}$ for some integer k , and so we conclude that $K = \mathbb{Q}(\sqrt{d})$. $\square$

The fact that Pell's Equation and quadratic fields can both be determined by a squarefree integers suggests that there might be some connection between the two. If we consider explicit representations of elements of a quadratic field K , we see that this is the case. Let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field for a squarefree integer d . Clearly $\sqrt{d}$ has minimal polynomial $x^2 - d$ over $\mathbb{Q}$, the other root of which is $-\sqrt{d}$, which also lies in K . Furthermore, we can see that $\{1, \sqrt{d}\}$ is a basis for K as a $\mathbb{Q}$ -vector space. This set is obviously linearly independent, and to see that this set spans K , notice that we have an isomorphism $K \cong \mathbb{Q}[x]/\langle x^2 - d \rangle$ given by mapping $\sqrt{d}$ to the congruence class of x . The linear independence of $\{1, \sqrt{d}\}$ allows us to represent $\alpha \in K$ as $\alpha = a + b\sqrt{d}$, where $a, b \in \mathbb{Q}$.

It is often desirable to consider the **ring of integers** of a number field K . The ring of integers is the subset R of K consisting of elements of K that satisfy polynomials over $\mathbb{Z}$. It is a standard result that the ring of integers for a quadratic field $\mathbb{Q}(\sqrt{d})$ is equal to $\mathbb{Z}[\sqrt{d}] = \{a + b\sqrt{d} \mid a, b \in \mathbb{Z}\}$ when $d \not\equiv 1 \pmod{4}$,

and is equal to $\mathbb{Z}[\frac{1+\sqrt{d}}{2}]$ when $d \equiv 1 \pmod{4}$. If K is a number field, and R is the ring of integers of K , then the group of units of K is the (multiplicative) group of invertible elements of R . The **norm** of an element of K is equal to

$$N(a + b\sqrt{d}) = (a + b\sqrt{d})(a - b\sqrt{d}) = a^2 - db^2.$$

The map $N : K \rightarrow \mathbb{R}$ is multiplicative, and so it must be that $N(a) = \pm 1$ for any unit a of K . Therefore units are precisely those elements of K such that

$$a^2 - db^2 = \pm 1,$$

which is Pell's Equation in the case where ± 1 is positive. We will see in Section 5 that the group of units for a quadratic field is greatly helpful in solving Pell's Equation.

3 Lattices

This section draws material from [3, §4.1].

We now turn our attention to the theory of lattices, which will provide us with results critical to analyzing the group of units for an arbitrary number field.

3.1 Discrete Subgroups and Lattices in $\mathbb{R}^n$

Consider $\mathbb{R}^n$ as an $\mathbb{R}$ -vector space. $\mathbb{R}^n$ is a group with respect to addition, and so we may consider subgroups of $\mathbb{R}^n$. We are particularly interested in a certain class of subgroups of $\mathbb{R}^n$.

Definition 1. A subgroup H of $\mathbb{R}^n$ is called **discrete** if $H \cap S$ is finite for every compact subset $S \subseteq \mathbb{R}^n$.

The Heine-Borel Theorem tells us that $S \subseteq \mathbb{R}^n$ is compact if and only if it is closed and bounded. However, there is an even more intuitive way of understanding the above definition. Throughout this section, let $B_r(x)$ denote the closed ball of radius $r \in \mathbb{R}$ about a point $x \in \mathbb{R}^n$.

Proposition 2. A subgroup H of $\mathbb{R}^n$ is discrete if and only if every infinite subset of H is unbounded.

Proof. First, let H be discrete, and let $A \subseteq H$ have infinite cardinality. We go by contradiction. Suppose that A is bounded. Then there exists some $r \in \mathbb{R}$ such that $A \subseteq B_r(0)$. Since $B_r(0)$ is compact, this contradicts the discreteness of H . Therefore every infinite subset of a discrete subgroup of $\mathbb{R}^n$ is unbounded.

Now let H be such that all of its infinite subsets are unbounded, and let S be a compact subset of $\mathbb{R}^n$. Since S is bounded, $S \cap H$ is bounded. Therefore $S \cap H$ is finite, so H is discrete. $\square$

A certain kind of discrete subgroup of $\mathbb{R}^n$ is of particular importance.

Definition 2. A **lattice** in $\mathbb{R}^n$ is a $\mathbb{Z}$ -module generated by some basis for $\mathbb{R}^n$ as a vector space.

Notice that a lattice is naturally an additive subgroup of $\mathbb{R}^n$. In fact, we have a sharper result.

Proposition 3. Every lattice Λ in $\mathbb{R}^n$ is a discrete subgroup of $\mathbb{R}^n$.

Proof. Let $v_1, \dots, v_n$ comprise a basis for $\mathbb{R}^n$ that generates Λ . We first show that Λ is a subgroup of $\mathbb{R}^n$. Let $u, w \in \Lambda$. Since Λ is a $\mathbb{Z}$ -module, u and w can be written

$$u = \sum_{i=1}^n a_i v_i, \quad w = \sum_{i=1}^n b_i v_i$$

such that $a_i, b_i \in \mathbb{Z}$ for $1 \leq i \leq n$. Then $u + w = \sum_{i=1}^n (a_i + b_i) v_i$. Since $a_i + b_i \in \mathbb{Z}$ for $1 \leq i \leq n$, we see that Λ is closed under addition. It is trivial to see that Λ contains an additive identity and additive inverses, and so Λ is a subgroup of $\mathbb{R}^n$.

We now show that Λ is in fact a discrete subgroup of $\mathbb{R}^n$. Since the v_i span $\mathbb{R}^n$, every element of $\mathbb{R}^n$ admits of a unique representation $a_1v_1 + \cdots + a_nv_n$ for $a_i \in \mathbb{R}$, $1 \leq i \leq n$. Therefore the map $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^n$ given by $\varphi(a_1v_1 + \cdots + a_nv_n) = (a_1, \dots, a_n)$ is well-defined. Since φ is a linear map, φ is continuous. Therefore if $S \subseteq \mathbb{R}^n$ is compact, then $\varphi(S)$ is also compact. There must then exist some ball $B_r(0)$ such that $\varphi(S) \subseteq B_r(0)$. Therefore $a_1v_1 + \cdots + a_nv_n \in S$ implies that $|(a_1, \dots, a_n)| \leq r$. Therefore $|a_i| \leq r$ for $1 \leq i \leq n$. Since Λ is a $\mathbb{Z}$ -module generated by the v_i , there can only be finitely many elements $a_1v_1 + \cdots + a_nv_n \in \Lambda$ such that $|a_i| \leq r$. Therefore there can only be finitely elements of Λ contained in S . We conclude that Λ is discrete. $\square$

The second part of the above proof is due to Jarvis [2]. We are now equipped to prove some fundamental theorems about lattices.

3.2 Minkowski's Theorem

For any lattice Λ in $\mathbb{R}^n$, there is an important subset of $\mathbb{R}^n$ determined by Λ :

Definition 3. Let Λ be generated by the linearly independent vectors $v_1, \dots, v_n \in \mathbb{R}^n$. We then define the **fundamental parallelepiped** of Λ to be the set

$$P_\Lambda = \left\{ x \in \mathbb{R}^n \mid x = \sum_{i=1}^n a_i v_i, 0 \leq a_i < 1 \right\}.$$

It is a basic result in linear algebra that the volume of P_Λ is equal to $|\det(v_1, \dots, v_n)|$. For a subset S of $\mathbb{R}^n$, we will denote the volume of S by $\mu(S)$. We will only ever consider subsets of $\mathbb{R}^n$ whose volume is well-defined.

Proposition 4. Let Λ be a lattice in $\mathbb{R}^n$ generated by $v_1, \dots, v_n$. Then every $x \in \mathbb{R}^n$ can be written $x = v + \tilde{x}$, where $v \in \Lambda$ and $\tilde{x} \in P_\Lambda$.

Proof. Since $v_1, \dots, v_n$ constitute a basis for $\mathbb{R}^n$, x admits of a unique representation as a linear combination $x = a_1v_1 + \cdots + a_nv_n$ for $a_i \in \mathbb{R}$, $1 \leq i \leq n$. Let $\lfloor a \rfloor$ denote the greatest integer less than a for any $a \in \mathbb{R}$. Then we have

$$x = \sum_{i=1}^n \lfloor a_i \rfloor v_i + \left(x - \sum_{i=1}^n \lfloor a_i \rfloor v_i \right).$$

Clearly $\sum_{i=1}^n \lfloor a_i \rfloor v_i \in \Lambda$, and

$$x - \sum_{i=1}^n \lfloor a_i \rfloor v_i = \sum_{i=1}^n (a_i - \lfloor a_i \rfloor) v_i.$$

Since each $a_i - \lfloor a_i \rfloor$ is in $[0, 1)$, we have that $x - \sum_{i=1}^n \lfloor a_i \rfloor v_i \in P_\Lambda$. This concludes our proof. $\square$

The following lemma will be significant to the proof of Minkowski's Theorem. For an element a of $\mathbb{R}$ and a subset S of $\mathbb{R}^n$, let $aS = \{x \in \mathbb{R}^n \mid x = as, s \in S\}$ and let $a + S = \{x \in \mathbb{R}^n \mid x = a + s, s \in S\}$.

Lemma 5. Let Λ be a lattice in $\mathbb{R}^n$ generated by $v_1, \dots, v_n$, and let $S \subseteq \mathbb{R}^n$ be such that $\mu(S) > \mu(P_\Lambda)$. Then there exist distinct $x, y \in S$ such that $x - y \in \Lambda$.

Proof. By the above proposition, it is clear that $\mathbb{R}^n = \bigsqcup_{v \in \Lambda} (v + P_\Lambda)$. Therefore $S = \bigsqcup_{v \in \Lambda} (S \cap (v + P_\Lambda))$, so that

$$\mu(S) = \sum_{v \in \Lambda} \mu(S \cap (v + P_\Lambda)).$$

The above sum is finite because S is bounded. Since the volume of a set is invariant under translation, we have that $\mu(S \cap (v + P_\Lambda)) = \mu((-v + S) \cap P_\Lambda)$ for every $v \in \Lambda$. If the sets $(-v + S) \cap P_\Lambda \subseteq P_\Lambda$ are all

pairwise disjoint, then $\sum_{v \in \Lambda} \mu((-v + S) \cap P_\Lambda) \leq \mu(P_\Lambda)$, contradicting our assumption that $\mu(S) > \mu(P_\Lambda)$. Therefore there must be two distinct $v, w \in \Lambda$ such that $(-v + S) \cap P_\Lambda$ and $(-w + S) \cap P_\Lambda$ are not disjoint, and so there are elements $x, y \in S$ such that $-v + x = -w + y$. Since v and w are distinct, x and y must be too. We now have $x - y = v - w \in \Lambda$, and so we are done. $\square$

We can now prove Minkowski's Theorem, a result central to the geometry of numbers.

Theorem 6 (Minkowski). *Let Λ be a lattice in $\mathbb{R}^n$ generated by $v_1, \dots, v_n$, and let S be a measurable subset of $\mathbb{R}^n$ such that*

- $x \in S \implies -x \in S$ (i.e. S is symmetric with respect to the origin), and
- $x, y \in S \implies \{tx + (1-t)y \mid t \in [0, 1]\} \subseteq S$ (i.e. S is convex).

If S satisfies either of the following:

- (i) $\mu(S) > 2^n \mu(P_\Lambda)$,
- (ii) $\mu(S) \geq 2^n \mu(P_\Lambda)$ and S is compact,

then S contains an element of Λ other than $\mathcal{O}$ (the origin).

Proof. We begin with case (i). Consider the set $\frac{1}{2}S$. We have $\mu(\frac{1}{2}S) = 2^{-n}\mu(S) > \mu(P_\Lambda)$, and so by Lemma 5 there exist distinct $x, y \in \frac{1}{2}S$ such that $x - y \in \Lambda$. Since $x, y \in \frac{1}{2}S$, we have $2x, 2y \in S$. Since S is symmetric about $\mathcal{O}$, we have $-2x \in S$. Since S is convex we have $\frac{1}{2}2x - \frac{1}{2}2y \in S$. Therefore $x - y \in S$, and so $x - y \in \Lambda$ is an element of S other than the origin.

Now we consider case (ii). Since $\mu(S) \geq 2^n \mu(P_\Lambda)$, we have that $\mu((1+\varepsilon)S) > 2^n \mu(P_\Lambda)$ for every $\varepsilon > 0$. Therefore $(1+\varepsilon)S$ satisfies the hypothesis of case (i), so that each $(1+\varepsilon)S$ contains some element of Λ other than $\mathcal{O}$. That is, $(\Lambda \setminus \{\mathcal{O}\}) \cap (1+\varepsilon)S$ is nonempty for every $\varepsilon > 0$. Denote $(\Lambda \setminus \{\mathcal{O}\}) \cap (1+\varepsilon)S$ by $S_{\Lambda, \varepsilon}$. Each $S_{\Lambda, \varepsilon}$ is finite (since Λ is a discrete subgroup of $\mathbb{R}^n$), and so each of these sets is compact. Furthermore, $S_{\Lambda, \varepsilon} \subseteq S_{\Lambda, \delta}$ whenever $\varepsilon < \delta$. Therefore $\{S_{\Lambda, \varepsilon} \mid \varepsilon > 0\}$ is a nested family of nonempty compact sets, and so

$$\bigcap_{\varepsilon > 0} S_{\Lambda, \varepsilon} \neq \emptyset.$$

Therefore there is some $v \in \Lambda \setminus \{\mathcal{O}\}$ contained in $(1+\varepsilon)S$ for every $\varepsilon > 0$. Therefore v is at most at distance ε from the set S for any ε , and so v must also be in S by the compactness of S . $\square$

As we will see in the next section, Minkowski's Theorem is an important tool for proving Dirichlet's Unit Theorem.

4 Dirichlet's Unit Theorem

This section draws material from [3, §§4.2, 4.4].

In this section we take the final steps to building up many of the notions relevant to Dirichlet's Unit Theorem.

4.1 Embeddings of Number Fields

Let K be a number field. Recall from Section 2.1 that K has a primitive element α such that $K = \mathbb{Q}(\alpha)$. Let f be the minimal polynomial of α , and let $n = \deg(f)$. We know that f splits into n linear factors over $\mathbb{C}$, so let $\alpha_1, \dots, \alpha_n \in \mathbb{C}$ be the roots of f , where $\alpha_1 = \alpha$. Then we have n injective field homomorphisms $\sigma_i : K \rightarrow \mathbb{C}$ that fix $\mathbb{Q}$, where $\sigma_i(\alpha) = \alpha_i$. Let r_1 be the number of these embeddings for which $\sigma_i(K) \subseteq \mathbb{R}$. Then $n - r_1$ is the number of these embeddings whose range is not contained in $\mathbb{R}$. Notice that if α_i is a root of f , then so is $\overline{\alpha_i}$, so that we have pairs of embeddings $\alpha \mapsto \alpha_i$ and $\alpha \mapsto \overline{\alpha_i}$ for each $\alpha_i \notin \mathbb{R}$. Therefore $n - r_1$ is even, so we write $n - r_1 = 2r_2$. Rearrange the indices i such that for $1 \leq i \leq r_1$ we have $\sigma_i(K) \subseteq \mathbb{R}$, and for $r_1 + 1 \leq i \leq r_1 + r_2$ we have that $\sigma_i(\alpha) = \alpha_i$ and $\sigma_{i+r_2}(\alpha) = \overline{\alpha_i}$. In a sense, the final r_2 embeddings are redundant, since they are determined by the previous r_2 embeddings. We are now equipped to discuss Dirichlet's Unit Theorem.

4.2 Dirichlet's Unit Theorem

Dirichlet's Unit Theorem is a powerful result concerning the group of units of a number field (recall Section 2.2). We will give an abbreviated sketch of the proof that highlights the connections of Dirichlet's Unit Theorem to the concepts discussed so far. The full proof is quite long and involves an even more sophisticated theory of number fields than is developed here. To see the full proof, refer to [3].

Theorem 7 (Dirichlet). *Let K be a number field of degree n over $\mathbb{Q}$, and let r_1 and r_2 be the integers defined in Section 4.1. The multiplicative group A of units of K is isomorphic to $\mathbb{Z}^{r_1+r_2-1} \times G$, where G is the cyclic group consisting of the roots of unity contained in K .*

Proof sketch. We first define the logarithmic embedding of a number field. Let K be a number field with embeddings σ_i into $\mathbb{C}$ as in Section 4.1. The **logarithmic embedding** L of K into $\mathbb{R}^{r_1+r_2}$ is given by

$$L(x) = (|\log(\sigma_1(x))|, |\log(\sigma_2(x))|, \dots, |\log(\sigma_{r_1+r_2}(x))|).$$

The logarithmic embedding L maps K into $\mathbb{R}^{r_1+r_2}$. We want to show that the image of the group of units A under L is a discrete subgroup of $\mathbb{R}^{r_1+r_2}$. Since L is a group homomorphism taking the multiplicative group A to the additive group $L(A)$, the image $L(A)$ is certainly a subgroup of $\mathbb{R}^{r_1+r_2}$. Now let B be a compact subset of $\mathbb{R}^{r_1+r_2}$. Since B is bounded, there exists some bound $M \in \mathbb{R}$ such that for every $x \in A$,

$$\begin{aligned} L(x) \in B &\implies -M \leq \log |\sigma_i(x)| \leq M, \quad 1 \leq i \leq r_1 + r_2 \\ &\implies e^{-M} \leq |\sigma_i(x)| \leq e^M, \quad 1 \leq i \leq r_1 + r_2. \end{aligned}$$

It is a standard result that the $\sigma_i(x)$'s determine the coefficients of a polynomial $g \in \mathbb{Z}[x]$ that is a power of the minimal polynomial for x (this g is in fact called the characteristic polynomial of x). Since the $\sigma_i(x)$'s are bounded, so are the coefficients of g . Since these coefficients are integers, it follows that there are only finitely many possibilities for the coefficients of a characteristic polynomial for an element x that maps into B . Therefore $(L|_A)^{-1}(B)$ is finite, so $L(A) \cap B$ must also be finite. Therefore $L(A)$ is a discrete subgroup of $\mathbb{R}^{r_1+r_2}$.

Let $\mathcal{O}$ represent the origin in $\mathbb{R}^{r_1+r_2}$. Then $\{\mathcal{O}\}$ is compact, and so $(L|_A)^{-1}(\{\mathcal{O}\})$, the kernel of $L|_A$, is finite. The only elements of A that map to $\mathcal{O}$ are the roots of unity in K , and these elements form a cyclic group G . Furthermore, $L(A)$ is a discrete subgroup of $\mathbb{R}^{r_1+r_2}$, and it is a basic result that there is therefore some subspace of $\mathbb{R}^{r_1+r_2}$ in which $L(A)$ is a lattice. Therefore $L(A)$ is isomorphic to $\mathbb{Z}^s$ for some power $s \leq r_1 + r_2$.

The proof that s is precisely $r_1 + r_2 - 1$ is beyond the scope of this paper, but Minkowski's Theorem plays a central role. $\square$

5 Solving Pell's Equation

This section draws material from [3, §4.6].

Dirichlet's Unit Theorem allows us to talk explicitly about the group of units in a number field, allowing us generate solutions to Pell's Equation when we look at quadratic fields.

5.1 Groups of Units in Real Quadratic Fields

Let $K = \mathbb{Q}(\sqrt{d})$ be a quadratic field, where d is squarefree and positive (i.e. $\sqrt{d} \in \mathbb{R}$). Since $\sqrt{d}$ and $-\sqrt{d}$ are the only roots of the minimal polynomial $x^2 - d$ of $\sqrt{d}$, we see that every embedding of K into $\mathbb{C}$ has a subset of $\mathbb{R}$ as its range (recall Section 4.1). Therefore $r_1 = 2$ and $r_2 = 0$, so that the group of units A of K is isomorphic to $\mathbb{Z} \times \{1, -1\}$ by Dirichlet's Unit Theorem. The positive units correspond to the subgroup $\mathbb{Z} \times \{1\} \cong \mathbb{Z}$, and so the group A^+ of positive units of K is cyclic of infinite order. A^+ has two generators, one greater than the identity (i.e. 1), and one less than the identity. We call the generator $u \in A^+$ that is greater than 1 the **fundamental unit** of K .

5.2 Generating Solutions to Pell's Equation

Recall (from Section 2.2) that the units of K are precisely those elements $a+b\sqrt{d}$ of K such that $N(a+b\sqrt{d}) = a^2 - b^2d = \pm 1$. It turns out that the fundamental unit of a number field K is capable of generating integer solutions to Pell's Equation in every case, but we restrict our attention to a particular, informative case.

Let u be the fundamental unit for a number field $K = \mathbb{Q}(\sqrt{d})$, and suppose both that $N(u) = 1$ and that $A = \mathbb{Z}[\sqrt{d}]$. Recall from Section 2.2 that A is of this form whenever $d \not\equiv 1 \pmod{4}$. Then $u = a_1 + b_1\sqrt{d}$ gives an integer solution (a_1, b_1) to Pell's Equation $x^2 - dy^2 = 1$. Furthermore, since u generates the multiplicative group A^+ , every positive unit is of the form $u^k = a_k + b_k\sqrt{d}$, $k \geq 0$, $a_k, b_k \in \mathbb{Z}$. Therefore each pair (a_k, b_k) is also a solution to Pell's Equation. This constitutes an algorithm for generating infinitely many solutions to Pell's Equation, the goal we had originally set out in Section 1.

Consider the following example. The fundamental unit for $\mathbb{Q}(\sqrt{7})$ is $8 + 3\sqrt{7}$, and we see that $7 \equiv 3 \not\equiv 1 \pmod{4}$. Observe that $8^2 - 7 \cdot 3^2 = 1$, and so we can generate positive units

$$(8 + 3\sqrt{7})^2 = 127 + 48\sqrt{7}, \quad (8 + 3\sqrt{7})^3 = 2024 + 765\sqrt{7}, \quad \dots$$

Therefore the pairs $(8, 3), (127, 48), (2024, 765), \dots$ are all solutions to Pell's Equation. Further, as we remarked in Section 1, we have approximations to $\sqrt{7}$:

$$\begin{aligned} \frac{8}{3} &\approx 2.66667 \\ \frac{127}{48} &\approx 2.64583 \\ \frac{2024}{765} &\approx 2.64575. \end{aligned}$$

Since $\sqrt{7} \approx 2.64575$, we see that our third solution $(2024, 765)$ already produces for us an approximation accurate to at least five places.

We have now seen the deep relationship between Pell's Equation, quadratic fields, and lattices in $\mathbb{R}^n$. There is a great deal more to be said about this theory, and references [2] and [3] contain plenty of information on where one can go from this point.

References

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